

# Iris Flower Classification using K-Nearest Neighbors(KNN)

## 1. Overview

This project demonstrates a supervised machine learning approach to classify Iris flower species into three categories: Setosa, Versicolor, and Virginica. The classification is performed using the K-Nearest Neighbors (KNN) algorithm based on four physical features: sepal length, sepal width, petal length, and petal width. Using the classic Scikit-learn Iris dataset, the model identifies patterns in sepal and petal dimensions to make accurate predictions.

## 2. Dataset Preparation & Model Architecture:

- **Dataset Composition**

The Iris dataset contains **150 samples** with four features:

- Sepal Length (cm)
- Sepal Width (cm)
- Petal Length (cm)
- Petal Width (cm)

The target labels are encoded as:

- 0: Setosa
- 1: Versicolor
- 2: Virginica.

- **Data Split:**

The data was divided into a training set and a testing set with a 30% test size (N=45 for testing, N=105 for training).

- **Model Selection:**

A K-Nearest Neighbors classifier was initialized with the hyperparameter  $k=3$ . This means the model identifies the three closest points in the feature space to determine the class of a new observation.

The model achieved high precision across all classes:

Overall Accuracy: 97.78%

### 3. Performance Analysis:

The model demonstrates high reliability, with nearly all test samples correctly identified. Based on the classification report, we can observe the following breakdown of performance:

```
KNN Accuracy (K=3): 97.78%
              precision    recall  f1-score   support

     0           1.00        1.00        1.00         14
     1           0.95        1.00        0.97         18
     2           1.00        0.92        0.96         13

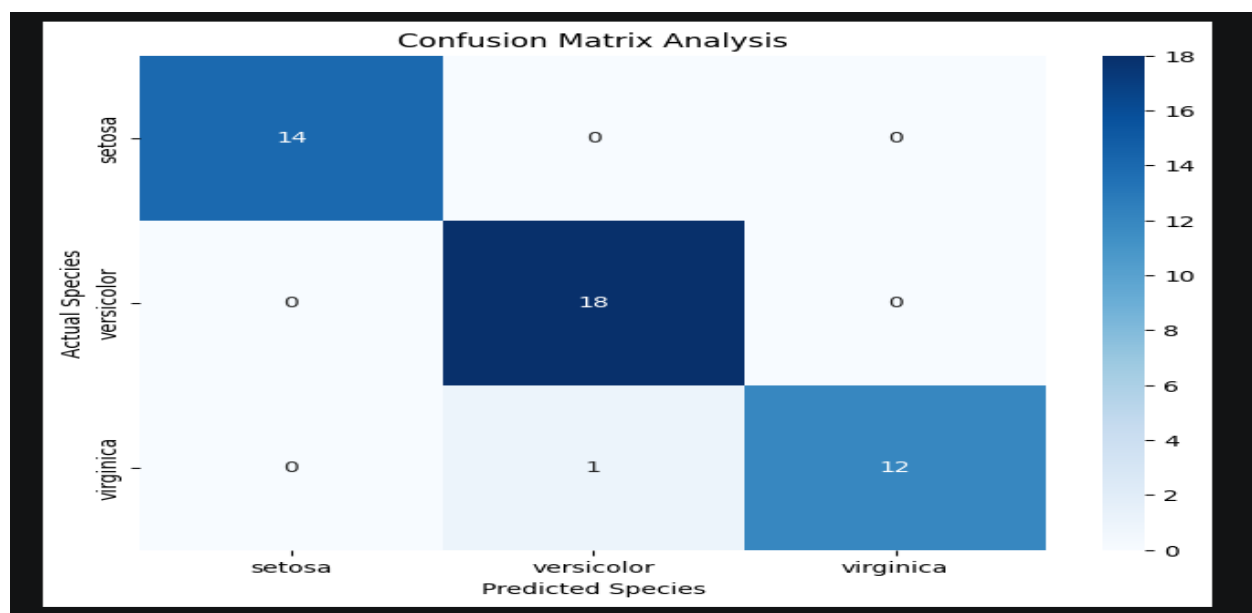
 accuracy                   0.98         45
 macro avg           0.98        0.97        0.98         45
 weighted avg           0.98        0.98        0.98         45
```

- Advance Performance Metrics:

| Advanced Performance Metrics: |           |          |          |         |
|-------------------------------|-----------|----------|----------|---------|
|                               | precision | recall   | f1-score | support |
| setosa                        | 1.000000  | 1.000000 | 1.000000 | 14.0    |
| versicolor                    | 0.947368  | 1.000000 | 0.972973 | 18.0    |
| virginica                     | 1.000000  | 0.923077 | 0.960000 | 13.0    |

- Confusion Matrix Analysis:

The confusion matrix provides a visual breakdown of the model's predictive accuracy across the testing set. It allows for granular visibility into true positive predictions and any specific misclassifications between the iris species.



## Interpretation of Results:

- **Diagonal Alignment:** The numerical values of 14 (Setosa), 18 (Versicolor), and 12 (Virginica) are concentrated along the primary diagonal, representing the "True Positives" where the model's prediction matched the actual label.
- **Misclassification Analysis:** A single instance of misclassification is observed in the Virginica class. The matrix shows that 1 sample of Iris virginica was incorrectly predicted as Iris versicolor.
- **Performance Summary:** Out of 45 total test samples, the model achieved a near-perfect classification rate. The visual gradient confirms that the KNN algorithm successfully established distinct boundaries for most of the dataset, with only slight overlap occurring between the Versicolor and Virginica features.

## 2. Project Conclusion:

The implementation of the  $k=3$  KNN model was highly successful, achieving an overall accuracy of 97.78%. While the Setosa species remains perfectly separable, the slight overlap between Versicolor and Virginica resulted in a single misclassification. Overall, the model proves to be an effective tool for this classification, providing high precision and recall across all categories.